

Gradient Descent by Hand: From Slopes to Fitting a Model

Name: _____

Partner(s): _____

Date: _____

Launch: the update rule

For an objective function $J(\theta)$, gradient descent uses

$$\theta^{(t+1)} = \theta^{(t)} - \alpha \nabla J(\theta^{(t)})$$

where $\alpha > 0$ is the learning rate.

1. Why is there a minus sign in this update?

2. Complete the sentence: the gradient points in the direction of steepest

_____, so $-\nabla J$ points in the direction of steepest
_____.

1. One parameter: follow the slope

Consider the objective

$$J(w) = w^2 - 6w + 11.$$

Here, the parameter space is the number line: $w \in \mathbb{R}$.

A. Understand the landscape

1. Compute the first and second derivatives.

$$J'(w) =$$

$$J''(w) =$$

2. Find the critical point by solving $J'(w) = 0$.

$$w^* =$$

3. Use $J''(w^*)$ to classify the critical point. Is it only a local minimum, or can you also conclude that it is the global minimum? Explain.

4. At $w = -1$, is $J'(w)$ positive or negative? In which direction should w move to decrease J ?

B. Run gradient descent

Use $w^{(0)} = -1$ and $\alpha = \frac{1}{4}$. Complete the table. The first row means

$$w^{(1)} = w^{(0)} - \alpha J'(w^{(0)}).$$

t	Current $w^{(t)}$	Gradient $J'(w^{(t)})$	New $w^{(t+1)}$	New loss $J(w^{(t+1)})$
0	-1			
1				
2				
3				

If the stopping rule is $|J'(w^{(t)})| < 0.6$, at which iterate do you stop?

$t =$

C. See what the learning rate does

Restart at $w^{(0)} = -1$. For each learning rate, apply the same gradient-descent update three times.

1. Complete the table.

Learning rate	$w^{(0)}$	$w^{(1)}$	$w^{(2)}$	$w^{(3)}$
$\alpha = \frac{1}{2}$	-1			
$\alpha = 1$	-1			

Use this space for calculations.

2. More generally, what do you observe about how the learning rate affects the size of the updates and the behavior of the sequence of iterates?

2. Two parameters: follow the gradient

Now consider

$$J(u, v) = (u - 1)^2 + (u - 1)(v + 1) + (v + 1)^2, \quad \boldsymbol{\theta} = \begin{bmatrix} u \\ v \end{bmatrix}.$$

Unlike a derivative in one dimension, the gradient is a vector in the two-dimensional parameter space.

A. Compute the gradient and Hessian

1. Differentiate with respect to each parameter.

$$\frac{\partial J}{\partial u} =$$

$$\frac{\partial J}{\partial v} =$$

2. Assemble the gradient and Hessian.

$$\nabla J(u, v) = \begin{bmatrix} \\ \end{bmatrix}, \quad \nabla^2 J(u, v) = \begin{bmatrix} & \\ & \end{bmatrix}.$$

3. Solve $\nabla J(u, v) = \mathbf{0}$ for the critical point.

$$\boldsymbol{\theta}^* = \begin{bmatrix} u^* \\ v^* \end{bmatrix} = \begin{bmatrix} \\ \end{bmatrix}.$$

B. Re-express the same objective and use its curvature

This part does **not** define a new loss. Let $a = u - 1$ and $b = v + 1$ be temporary shorthand for the shifted coordinates already present in the objective from

Part A. Substituting them into that same objective gives

$$\begin{aligned} J(u, v) &= a^2 + ab + b^2 \\ &= \frac{1}{2}a^2 + \frac{1}{2}b^2 + \frac{1}{2}(a + b)^2. \end{aligned}$$

1. Verify the final equality by expanding its right-hand side.

2. What does this representation of the original objective tell you about $J(u, v)$ when $(a, b) \neq (0, 0)$? Use it to explain why the Hessian is positive definite and why the critical point from Part A is the unique global minimum.

C. Run gradient descent in two dimensions

Use

$$\boldsymbol{\theta}^{(0)} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad \alpha = \frac{1}{4}.$$

Carry out two complete update steps. Both coordinates must be updated using the gradient evaluated at the same current point.

Update 1: from $\boldsymbol{\theta}^{(0)}$ to $\boldsymbol{\theta}^{(1)}$

1. Substitute $\boldsymbol{\theta}^{(0)}$ into your gradient from Part A.

$$\nabla J(\boldsymbol{\theta}^{(0)}) =$$

2. Apply the update rule and show both coordinate calculations.

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \alpha \nabla J(\boldsymbol{\theta}^{(0)}) =$$

3. Evaluate the loss at the new point.

$$J(\boldsymbol{\theta}^{(1)}) =$$

Update 2: from $\theta^{(1)}$ to $\theta^{(2)}$

1. Recompute the gradient at the new current point. Do not reuse the previous gradient.

$$\nabla J(\theta^{(1)}) =$$

2. Apply the update rule and show both coordinate calculations.

$$\theta^{(2)} = \theta^{(1)} - \alpha \nabla J(\theta^{(1)}) =$$

3. Evaluate the loss at the new point.

$$J(\boldsymbol{\theta}^{(2)}) =$$

4. Compare $J(\boldsymbol{\theta}^{(0)})$, $J(\boldsymbol{\theta}^{(1)})$, and $J(\boldsymbol{\theta}^{(2)})$. What do you observe?

3. Fit a line in parameter space

Suppose we observe the following training data:

Observation i	Feature x_i	Response y_i
1	-1	-1
2	0	1
3	1	3

We will fit the linear model

$$\hat{y}_i = \beta_0 + \beta_1 x_i.$$

The model has two parameters, so a candidate model is a point

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} \in \mathbb{R}^2.$$

Use the half-mean-squared-error objective

$$J(\boldsymbol{\beta}) = \frac{1}{2n} \sum_{i=1}^n (\beta_0 + \beta_1 x_i - y_i)^2, \quad n = 3.$$

The factor $1/2$ is included only to cancel the 2 produced by differentiating a square. It does not change the minimizing parameter values.

A. Translate model fitting into optimization

1. What does one point (β_0, β_1) in parameter space represent?

2. What does one point (x_i, y_i) in data space represent?

3. Define the residual using the prediction-minus-observation convention:

$$r_i(\boldsymbol{\beta}) = \hat{y}_i - y_i =$$

B. Build the gradient from the observations

Apply the chain rule to one term of the loss, then sum over the observations.
Show your derivation in the space below.

Your result should simplify to

$$\nabla J(\boldsymbol{\beta}) = \frac{1}{n} \sum_{i=1}^n r_i(\boldsymbol{\beta}) \begin{bmatrix} 1 \\ x_i \end{bmatrix}$$

Complete the two coordinates:

$$\frac{\partial J}{\partial \beta_0} = \frac{1}{3} \sum_{i=1}^3$$

,

$$\frac{\partial J}{\partial \beta_1} = \frac{1}{3} \sum_{i=1}^3$$

.

In words, each observation contributes a vector to the gradient. The feature x_i controls how much that observation contributes to the slope coordinate.

C. Compute the first full-data gradient

Start with $\beta^{(0)} = [0, 0]^T$. Complete the table and then add the three observation-level gradient contributions.

i	x_i	y_i	$\hat{y}_i$	$r_i = \hat{y}_i - y_i$	$r_i \begin{bmatrix} 1 \\ x_i \end{bmatrix}$
1	-1	-1			
2	0	1			
3	1	3			
Sum					

Therefore,

$$\nabla J(\beta^{(0)}) = \frac{1}{3} \begin{bmatrix} & \end{bmatrix} = \begin{bmatrix} & \end{bmatrix}.$$

D. Run full-batch gradient descent

Use $\alpha = \frac{3}{4}$ and complete one full update. After finding the new parameter vector, compute its predictions and residuals to evaluate the new loss.

t	Current $\beta^{(t)}$	$\sum r_i$	$\sum x_i r_i$	$\nabla J(\beta^{(t)})$	New $\beta^{(t+1)}$	New loss $J(\beta^{(t+1)})$
0	$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$					

Show the parameter update in detail.

At $\beta^{(1)}$, compute the new predictions and residuals.

i	x_i	y_i	$\hat{y}_i$	$r_i = \hat{y}_i - y_i$
1	-1	-1		
2	0	1		
3	1	3		

Use the new residuals to compute $J(\beta^{(1)})$.

1. From $t = 0$ to $t = 1$, did β_0 and β_1 move through the data space or the parameter space?

2. What happened to the loss after the update?

3. Translate $\beta^{(1)}$ back into a fitted line:

$$\hat{y} = \quad + \quad x.$$

E. Identify and verify the optimum

1. Find the line that passes through all three observations.

$$\beta^* = \begin{bmatrix} \quad \\ \quad \end{bmatrix}.$$

2. Evaluate $\nabla J(\beta^*)$. What should you get?

3. For this dataset, the Hessian is

$$\nabla^2 J(\beta) = \frac{1}{3} \sum_{i=1}^3 \begin{bmatrix} 1 & x_i \\ x_i & x_i^2 \end{bmatrix}.$$

Compute it.

$$\nabla^2 J(\boldsymbol{\beta}) = \begin{bmatrix} & \\ & \end{bmatrix}.$$

4. Is this Hessian positive definite? What does that tell you about the critical point and the shape of the loss in parameter space?